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1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

a) Identify the key differences between MOSFET and BJT devices with respect to analogue circuit design. Describe when you would use one instead of the other using examples where appropriate.

- MOSFETs can be considered to be transconductance amplifiers exhibiting a square law relationship between voltage in and current out. They are majority carrier devices. BJTs however provide a constant, fixed current gain (beta), in addition to an exponential transconductance relationship (between voltage in and current out). BJTs are minority carrier devices. [1]

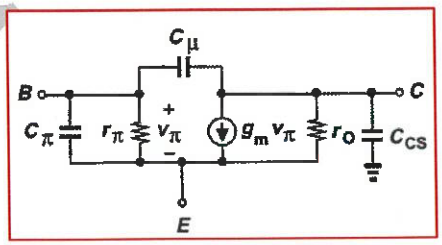
- In discrete (PCB-level) design, both BJT and MOSFET devices are typically available as 3-terminal devices (MOSFETs typically body connected to source terminal). BJTs require an input (base) current, whereas MOSFETs don't. [1]

- As a consequence, the input impedance of MOSFET devices is significantly higher than BJT devices. The vast majority of discrete transistors available are BJT devices (typically MOSFETs are used more as power transistors). This is partially due to MOSFET devices being sensitive to electrostatic discharge (ESD) damage. [1]

- In integrated design, the most abundant technology is CMOS that generally only provides MOSFET devices (NMOS and PMOS). As a consequence, analogue design in the majority of mixed-signal integrated circuits utilize only MOSFETs. [1]

- One exception in CMOS technology where BJTs are typically used (using parasitic structures) is for bandgap reference circuits. Here the device physics are exploited to provide a well-defined temperature dependence, with little or no current gain. [1]

b) Draw the *small signal model* of an integrated BJT including parasitic capacitances and explain what limits its high frequency performance when used as a common emitter amplifier. [5]



- SS model from (b) above [3] including C_{CS} otherwise -1.

- High frequency response when configured as a CE amplifier is limited by Cμ that appears as a floating impedance (capacitance) across the amplifier- since B is input and C is output. [1]

- Since it is an inverting amplifier, through miller effect this capacitance is "multiplied" by approximately the voltage gain to appear as an equivalent

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grounded capacitance at the input. It is this capacitance that limits high frequency response. [1]

c) Derive expressions (by inspection) for the voltage gains of the amplifier circuits shown below in Fig. 1.1. Include the effect of a finite output resistance. [10]

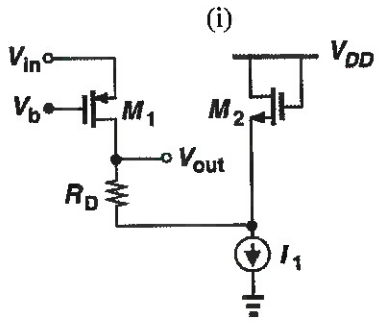


Fig. 1.1(a)

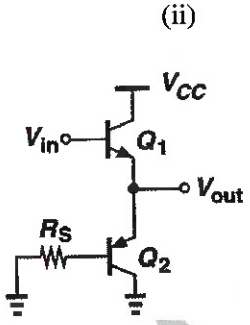


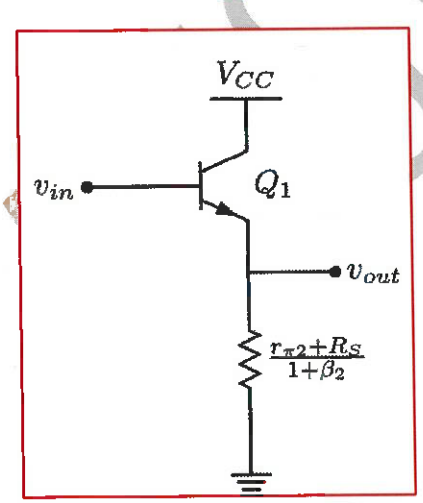
Fig. 1.1(b)

(i) Identify CG amplifier [1]. For this problem, recall that looking into the drain of a transistor with a grounded gate and source we see a resistance of r_o, and looking into either terminal of a diode-connected transistor we see a resistance of $1/g_m \parallel r_o$. [1]

$$A_v = g_{m1}\left(R_D + \frac{1}{g_{m2}}\right)$$

[3]

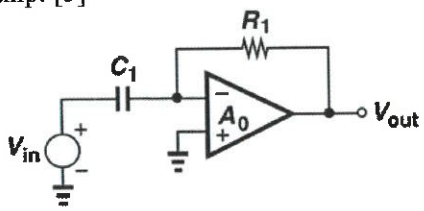
(ii) Identify EF [1] and to model Q2/RS as impedance of rpi2+Rs/(1+beta2), see schematic below. [2]



$$A_v = \frac{\frac{r_{\pi 2}+R_S}{1+\beta_2}}{\frac{1}{g_{m1}} + \frac{r_{\pi 2}+R_S}{1+\beta_2}}$$

[2]

d) We wish to design the differentiator of Fig. 1.2 for a pole frequency of 434MHz. If the values of R_1 and C_1 cannot be lower than 15kΩ and 2.2nF, respectively, compute the required gain of the op amp. [5]



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Fig. 1.2

In contrast to the ideal differentiator, the circuit contains a pole at

$$s_p = -\frac{A_0 + 1}{R_1 C_1}.$$

[2]

- 2 x pi x 434 x 10^6 = A_0 + 1/(15 x 10^3 * 2.2n) [2]
- A_0 = ~89986 [1]

e) Assuming perfect symmetry and $VA < \infty$, compute the differential voltage gain of the amplifier shown in Fig. 1.3. [5]

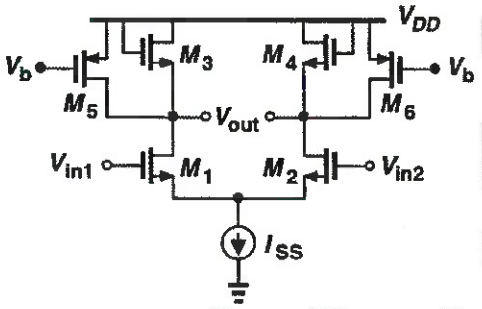
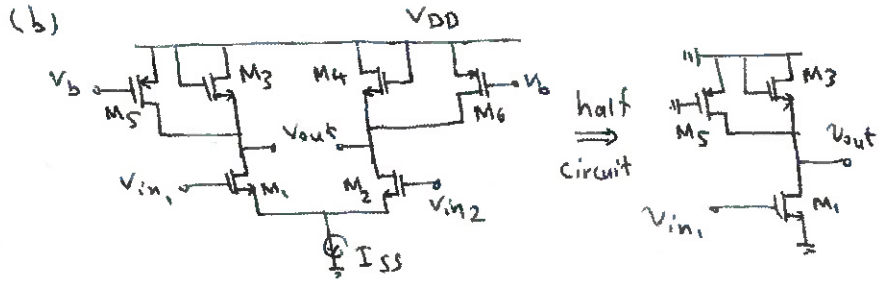


Fig. 1.3



$$A_v = -g_{m_1}\left(r_{o_1} \| r_{o_5} \| \frac{1}{g_{m_3}} \| r_{o_3}\right)$$

- Half circuit (see above) [2]
- Expression (see above) [2]
- Correct assumptions (r0x>>1/gmx), simplifying to -gm1 (r01||r05||1/gm3) [1]

f) The amplifier shown below in Fig. 1.4 consists of a common-gate stage (M_1 and R_D) and a feedback network (R_1, R_2, and M_2). Assuming R_1 and R_2 is very large and λ=0, compute the closed-loop gain and I/O impedances. [10]

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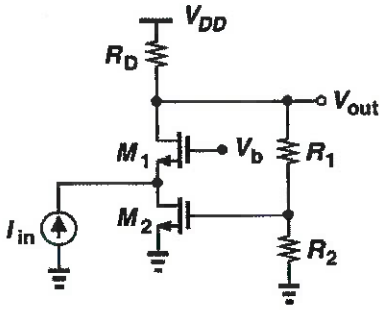


Fig. 1.4

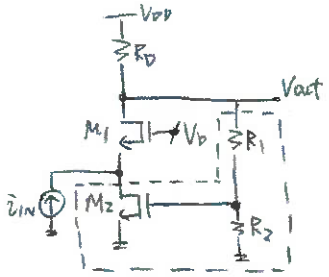
33.

$R_{O.L.} = \frac{V_{out}}{i_{IN}}$ (no feedback)

$= R_D$

K (feedback factor)

$= g_{m_2} \times \frac{R_2}{R_1 + R_2}$



$\lambda = 0$

$R_1 + R_2 \gg 1.$

$$\Rightarrow R_{C.L.} = \frac{V_{out}}{i_{IN}} = \frac{R_D}{1 + R_D \times g_{m_2} \frac{R_2}{R_1 + R_2}}$$

$$r_{in}|_{C.L.} = \frac{1/g_{m_1}}{1 + R_D \times g_{m_2} \frac{R_2}{R_1 + R_2}}$$

$$r_{out}|_{C.L.} = \frac{R_D}{1 + R_D \times g_{m_2} \frac{R_2}{R_1 + R_2}}$$

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- See above. R_{OL} [2], K[2], R_{CL}[2], RinCL [2] RoutCL [2]

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2. The circuit shown below in Fig. 3.1 is a two-stage amplifier consisting of a common source amplifier (M_1) followed by a common gate (M_2) stage.

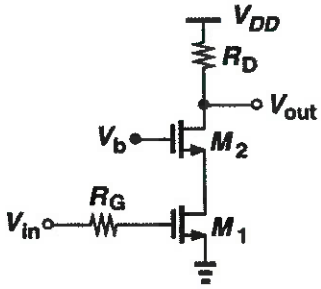


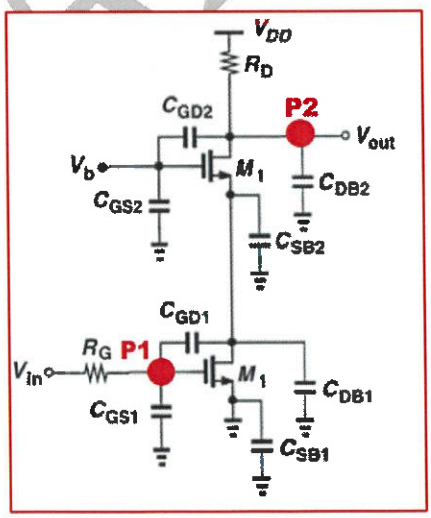
Fig. 3.1

Use the following parameters: $V_{DD} = 1.8V$, $(W/L)_1 = 100\mu m/0.18\mu m$, $(W/L)_2 = 50\mu m/0.18\mu m$, $I_{D1} = 2.5mA$, $\mu_n C_{ox} = 200\mu A/V^2$, $\lambda = 0$, $C_{ox} = 10fF/\mu m^2$, $C_{GS0}=C_{GS0}=C_0W$ where $C_0 = 0.2fF/\mu m$ denotes the gate-drain capacitance per unit width, $C_{DB}=C_{SB}=0$

a) Compare the frequency responses of a common source and common gate amplifier. [5]

- Common Source (CS) and Common Gate (CG) amplifiers have similar low frequency voltage gain (g_mR_D) except that CS is inverting and CG is non-inverting. [1]
- However, CG amplifier has better high frequency response than the CS due to the parasitic capacitances. [1]
- In case of the CS amplifier, the gate drain capacitance appears across the input/output terminal and is therefore subject to Miller multiplication (at the input). This significantly impacts the amplifier bandwidth by lowering the frequency of the input pole (since the input-referred capacitance is C_{GD} multiplied by the gain). [2]
- In the case of the CG amplifier since the gate is at AC ground, there is no Miller effect. [1]

b) Redraw the circuit above to: (i) identify any nodes that are associated with poles and (ii) include all the parasitic device capacitances (within M1-2). [5]

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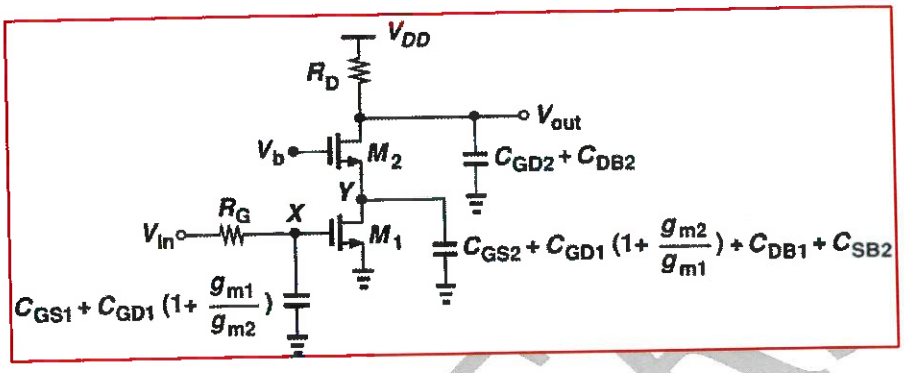
[4]

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- Note, node between CS and CG amplifiers may also be identified but must be discarded due to low impedance. [1] (or [1] for full expression but extra work to achieve).

c) Redraw the circuit drawn in your answer to (b) to remove any redundant capacitances and lump together any parallel capacitances. Determine expressions for the different pole frequencies using Miller's theorem where appropriate. [5]



- [3]

- $F_{p1}=1/[2\pi*R_G*(C_{GS1}+C_{GD1}(1+g_{m1}/g_{m2}))]$ [1]

- $F_{p2}=1/[2\pi*R_D*(C_{DB2}+C_{GD2})]$ [1]

Determine an expression for the DC (low frequency) gain of the amplifier and evaluate for R_D =2kΩ. [5]

- $A_v=A_{v1}*A_{v2} = -(g_{m1}/g_{m2}) * (g_{m2}R_D) = -g_{m1}R_D$ [3]

- $g_{m1}=\sqrt{(2I_{D1}(W/L)_2*\mu_nC_{ox})} = \sqrt{(2*2.5m*100/0.18*200\mu)} = 23.6mS$ [1]

- $A_V = -23.6mS * 2000 = -47.1$ [1]

d) Evaluate the DC gain and pole frequencies, for R_G=10kΩ. [5]

- $C_{GS1} = C_0W_1 + (2/3) * (W_1L_1) * C_{ox}$
  $= 0.2fF/\mu m * 100\mu m + (2/3) * (100 * 0.18) * 10f = 20fF + 120fF=140fF$ [1]

- $C_{GD1} = C_0W_1 = 0.2fF/\mu m * 100\mu m = 20fF$ and
  $C_{GD2} = C_0W_2 = 0.2fF/\mu m * 50\mu m = 10fF$ [1]

- $g_{m2} = \sqrt{(2I_{D1}(W_2/L_2)*\mu_nC_{ox})} = \sqrt{(2*2.5m*50/0.18*200\mu)} = 16.6mS$ [1]

- $F_{p1} = 1/[2\pi*R_G*(C_{GS1}+C_{GD1}(1+A_{v1}))]$
  $= 1/[2\pi*10K*(140f+20f(1+23.6/16.6))] = 84.5MHz$ [1]

- $F_{p2} = 1/[2\pi*R_D*(C_{DB2}+C_{GD2})] = 1/[2\pi*2K*(0+10f)] = 7.96GHz$ [1]

e) Discuss how the high frequency performance can be improved. [5]

- From the numeric results for F_{p1} and F_{p2} it is clear that F_{p1} is the dominant (relatively low frequency) pole. [1]

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- Looking at the expression this is directly proportional to R_G, so if R_G can be decreased this will increase F_{p1} by same factor. [1]
- Alternatively, the overall node capacitance at node X can also be reduced.[1]
- Majority is C_{GS1} that has a component that is proportional to W_1 and another component proportional to W_1*L_1. So C_{GS1} is directly proportional to W_1.[1]
- However, this also impacts the transconductance g_{m1}. So could trade-off power (i.e. increase $I_{D1}=I_{D2}$) to reduce W_1, this increasing f_{p1}. [1]

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3. The circuit shown below in Fig. 3.1 is an integrated current reference circuit.

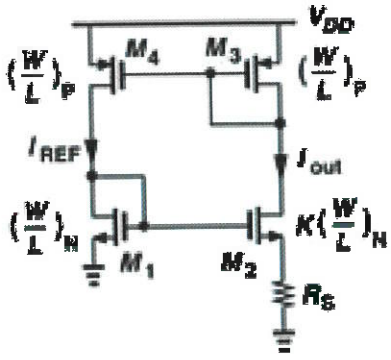


Fig. 3.1

Use the following parameters if needed: $V_{DD} = 1.8V$, $\mu_n C_{ox} = 200\mu A/V^2$, $\mu_p C_{ox} = 100\mu A/V^2$, $\lambda_n=0.1$, $\lambda_p=0.2$

a) Discuss the challenges (i.e. non-idealities) of designing a current reference circuit – both for a circuit board implementation, and within an integrated circuit. [5]

- Generally the three key issues are power supply sensitivity, process sensitivty (also mismatch) and temperature sensitivity. [3]
- Sensitivity to power supply affects both PCB and IC implementations for any currents or voltages that are derived from a power supply – for example potential divider. [1]
- In an integrated circuit, mismatch can be good - as everything on the same piece of silicon however all integrated components have temperature dependancies which are different. [1]

b) Which of these challenges (in your answer to part (a)), if any, does the circuit shown in Fig. 3.1 address? How can the remaining challenges be addressed? [5]

- The circuit above is insensitive to a first order to power supply (assuming good matching). [1]
- In order to arrive at a less sensitive solution, we postulate that the circuit must bias *itself*, i.e., I_{REF} must be somehow derived from I_{out}. The idea is that if I_{out} is to be ultimately independent of V_{DD}, then I_{REF} can be a replica of I_{out}. Fig. 11.2 illustrates an implementation where M_3 and M_4 copy I_{out}, thereby defining I_{REF}. In essence, I_{REF} is "bootstrapped" to I_{out}. With the sizes chosen here, we have $I_{out} = K I_{REF}$ if channel-length modulation is neglected. Note that, since each diode-connected device feeds from a current source, I_{out} and I_{REF} are relatively independent of V_{DD}. [1]
- Temperature sensitivity can be addressed by using a bandgap type reference. This generates a reference voltage/current that is the sum of two quantities with opposite temperature coefficients. [2]
- Sensitivity to process variation can be addressed by using trimming (or digital calibration), for example for the resistor value. [1]

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c) Derive an expression for the output current, I_{out}, stating any assumptions made. Hint: this needs to be a function of $(W/L)_N$ and R_S. [5]

require that $I_{out} = I_{REF}$ because they have identical dimensions. We can write $V_{GS1} = V_{GS2} + I_{D2}R_S$, or

$$\sqrt{\frac{2I_{out}}{\mu_n C_{ox}(W/L)_N}} + V_{TH1} = \sqrt{\frac{2I_{out}}{\mu_n C_{ox} K(W/L)_N}} + V_{TH2} + I_{out}R_S. \qquad (11.2)$$

Neglecting body effect, we have

$$\sqrt{\frac{2I_{out}}{\mu_n C_{ox}(W/L)_N}}\left(1 - \frac{1}{\sqrt{K}}\right) = I_{out}R_S, \qquad (11.3)$$

and hence

$$I_{out} = \frac{2}{\mu_n C_{ox}(W/L)_N} \cdot \frac{1}{R_S^2}\left(1 - \frac{1}{\sqrt{K}}\right)^2. \qquad (11.4)$$

As expected, the current is independent of the supply voltage (but still a function of process and temperature).

- See above. [1] per step

d) Design the circuit (i.e. determine the component values) such that an output current of I_{out}=10μA is generated. [5]

- 3 unknowns $(W/L)_N$, K and R_S. Need to start by selecting some values [1].
- K = 4 [1]
- W/L = 5 [1]
- This gives Rs = 10k [1]
- [1] if above values reasonable/feasible.

e) It is required to extend the circuit to provide a secondary output current of I_{out2}=250μA.

(i) Describe how the circuit could be extended to generate the secondary output. [3]

- Need to scale device M_3 to give 25x current gain. So if $(W/L)M_3=(W/L)M_4=1$, then $(W/L)_5=25$. [1]

  Probably too high to scale using a single mirror so can use two mirrors (M3/M4 to M5, with M5 sourcing current to a NMOS mirror with M6/M7). Therefore $(W/L)_5=5$, $(W/L)_6=1$, $(W/L)_6=5$ [2]

(ii) Calculate the total power consumption of the revised circuit. [2]

- Power = V_{DD} x (2 x 10μA (in current circuit) + 50μA (intemediate mirror) + 250μA (secondary output)) = 576μW [2]

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f) Explain why the configuration shown in Fig. 3.2 may be preferable to that in Fig. 3.1. [5]

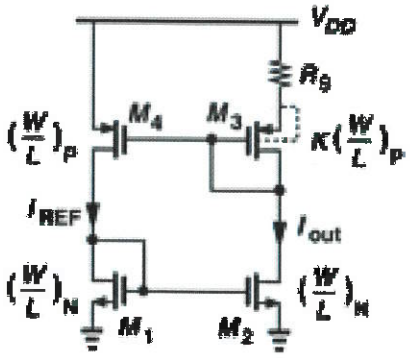


Fig. 3.2

- Since the source of device M2 in the original configuration is not grounded the threshold voltage of the NMOS device is dependant on VSB bias. This is called the body effect. [2]

- This can be addressed by "flipping" the entire circuit round. By using a PMOS device for the device that connects to the resistor the source can be connected to the bulk thus eliminating the body effect. [1]

- PMOS devices can be connected in this way as they can be formed in an independent N-well and therefore the body contact can be connected to any potential (above VSS) whereas for NMOS devices they typically all share the same substrate (so cannot bias at above VSS). [2]

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